Nagar Yuwak Shikshan Sanstha's
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Department of Computer Technology

Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering:

- Dynamic learning environment that encourages problem-solving using emerging technology
- National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry–institute collaboration
- Holistic, ethical and sustainable development for lifelong success.

Session	**2025-26 (EVEN)**	**Course Name**	**Operating Systems Lab**
Semester	**4**	**Course Code**	**23CT1402**
Roll No	**47**	**Name of Student**	**Dinesh More**

Practical Number	**02**
Course Outcome	1. Understand Computer System Configuration and Simulate system resources efficiently using Linux Commands (CO1) 2. Analyse operating system functionalities utilizing system calls, thread programming and process scheduling algorithms (CO2) 3. Apply Synchronization primitives to implement a Deadlock-free solution(CO3) 4. Simulate Disk scheduling, Memory allocation, File allocation, page replacement algorithms (CO4)
Aim	Aim: Execute Linux process management system calls.
Problem Definition	Execute Linux process management system calls.
Theory (100 words)	Theory:The Linux process management system calls are essential mechanisms provided by the operating system to create, control, and terminate processes. A process is an instance of a running program that includes its own memory space, execution state, and system resources. Linux follows a multitasking environment, allowing multiple processes to run concurrently. System calls such as fork() are used to create a new process by duplicating the calling process, while exec() replaces the current process image with a new program. The wait() and waitpid() system calls enable a parent process to wait for the completion of its child processes, ensuring proper synchronization. Process termination is handled using the exit() system call, which

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	releases all allocated resources. Additionally, system calls like getpid() and getppid() help in identifying processes uniquely. Understanding Linux process management system calls is crucial for efficient resource utilization, inter-process coordination, and the development of reliable and high-performance operating systems and applications.
Procedure and Execution (100 Words)	Procedure Open the Linux terminal or Ubuntu using WSL in Windows. Create a C program file using the Vim editor by typing vim process.c. Write a C program that uses Linux process management system calls such as fork, getpid, getppid, wait, and exit. Save the file and exit the editor. Compile the program using the GCC compiler by typing gcc process.c -o process. Execute the program by typing ./process. Observe the output showing the creation and execution of parent and child processes. Execution The program begins execution as a parent process. The fork system call creates a child process. Both the parent and child processes run concurrently. The child process displays its process ID and parent process ID. The parent process waits for the child process to complete using the wait system call.

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After the child process terminates using exit, the parent process resumes execution.
The successful execution confirms proper Linux process management.

Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    pid_t pid;
    int status;

    pid = fork();

    if (pid == 0) {
        // Child process
        printf("Child process running\n");
        exit(0);   // change to exit(1) or abort() for abnormal termination
    }
    else if (pid > 0) {
        // Parent process
        wait(&status);

        if (WIFEXITED(status)) {
            printf("Child terminated normally\n");
            printf("Exit status: %d\n", WEXITSTATUS(status));
        }
        else {
            printf("Child terminated abnormally\n");
        }
    }
    else {
```

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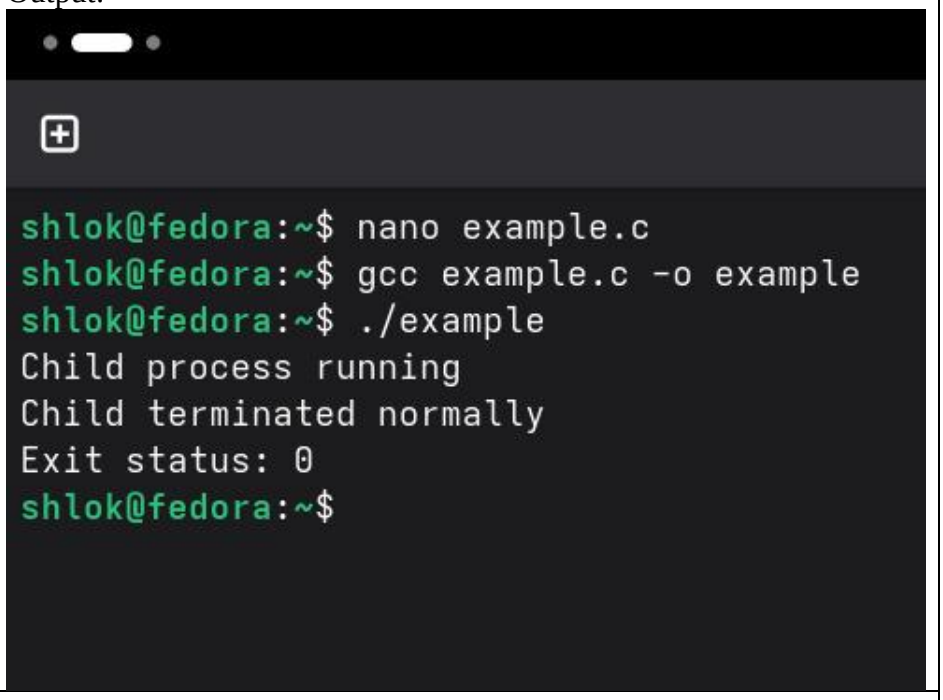
	printf("Fork failed\n");
	}
	
	return 0;
	}

	Output: 
Output Analysis	
Link of student Github profile where lab assignment has been uploaded	https://github.com/More-Dinesh
Conclusion	Understood process management system calls in Linux
Plagiarism Report (Similarity index	

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< 12%)	
Date	4th February 2026